

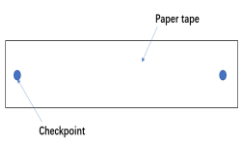
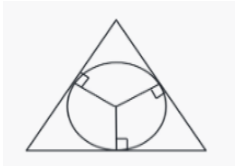
EXP1-1 Round peg in a square hole report

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Section1-Introduction: c

To measure the diameter of the three balls, there are three problems that needed to be solved. First, the surface of the balls is curved, whereas the ruler is straight and cannot be directly used to measure the diameter of the balls. The second problem arises at the same time: what methods should be chosen to measure the diameter of the balls? Then, the feasibility of the method remains to be verified.

Section2-Method:

Method	Schematic	Basketball	Tennis	Marble	Reason	Description
Overlapping paper method			✓	✓	Material is simple and easy to measure	1. Regularly stack the papers on one side 2. Fix the ball and adjust the paper until it is about the same height. 3. Take out the ruler and place it on top of the ball and the paper, making sure the ruler is parallel to the table. 4. Use the ruler to measure the thickness of the stacked paper.
Paper tape method		✓	✓		Portable and relatively accurate	1. Mark both ends of the ball's axis. 2. Cut the paper into tapes with parallel sides and straighten them. 3. Let the tape fit tightly to the surface of the sphere and make the two ends of the tape fit the mark on the ball.
Paper Clamp Ball Method		✓		✓	Scientific, simplified models	1. Hold three sheets upright and sandwich the ball in the middle, adjusting the length of the paper so that the top view is an equilateral triangle with a tangent ball. 2. Measure the length of paper l 3. Use the mathematical theorem relating to the inner tangent sphere to calculate the length of the sphere's diameter: $d=2r=2\sqrt{3}l/3$

Section3-Discussion:

The diameter range:

Basketball	Tennis	Marble
21.6cm-24.1cm	6.54–6.86 cm	0.5cm-2cm

1.Overlapping paper method

Advantages: This method is simple to operate in practice and the overlapping paper is portable and can be regularly stacked. Also, this method can be achieved by only one person, which can be used for marble.

Disadvantages: This method takes a lot of paper and is harder to measure for larger spheres, the amount of paper will increase dramatically when measure the diameter of tennis ball.

2.Paper tape method

Advantages: This is a measurement method suitable for medium and large spheres, which is easy to operate and can be calculated more conveniently. Moreover, it costs less than Overlapping paper method.

Disadvantages: This method may encounter situations where it does not work for small balls, which is difficult to accurately locate the sphere axis with paper tape, and where it is difficult to determine the diameter.

3.Paper Clamp Ball Method

Advantages: This method outperforms the other two in terms of science and precision. Using geometric and mathematical tools, this model can be extended to a larger field of research.

Disadvantages: It needs more mathematical and theoretical calculation, and cannot be measured directly to get the diameter size. Moreover, this method is more suitable for model calculation under ideal conditions, and has certain requirements on time and cost

Summary: Overlapping paper method is suitable for marble and other small size balls while Paper tape method fit larger size balls. Paper Clamp Ball Method can get accurate diameter but it needs ideal situation.